

Chapter 3.1 - The Balance of Payments

Professor Barker

2026-03-01

Table of contents

1	Introduction	3
	Context Setting	3
	Why the Balance of Payments Matters	4
	Chapter Roadmap	4
2	Fundamentals of BOP Accounting	4
2.1	What IS the Balance of Payments?	4
2.1.1	Key Clarifications	5
2.2	Common Misconceptions About the BOP	5
2.2.1	Misconception #1: “The BOP is a Balance Sheet”	5
2.2.2	Misconception #2: “The BOP Can Be in Disequilibrium”	5
2.2.3	Misconception #3: “Debits and Credits Work Like Corporate Accounting”	5
2.3	The BOP as a Flow Statement	5
2.3.1	1: Exchange of Real Assets	6
2.3.2	2: Exchange of Financial Assets	6
2.3.3	The “Follow the Cash Flow” Rule	6
2.4	The BOP Identity	7
2.4.1	Why the Need for “Net Errors and Omissions”?	7
3	The Accounts of the Balance of Payments	7
	Overview of the Main Accounts	7
3.1	The Current Account	7
3.1.1	The Four Components of the Current Account	8
3.1.2	Putting It Together: The Current Account Balance	10
3.2	The Capital Account	10
3.3	The Financial Account	11
3.3.1	The Three Main Components	11
3.3.2	The Fundamental Relationship: Current Account Financial Account	13
3.3.3	Breaking the Rules: China’s Twin Surpluses	14
3.4	Net Errors and Omissions	15
3.5	Official Reserves	15

4	BOP Impacts on Key Macroeconomic Rates	16
4.1	The BOP and Exchange Rates	16
4.1.1	The Simplified BOP Equation	16
4.1.2	Fixed Exchange Rate Countries	16
4.1.3	Floating Exchange Rate Countries	17
4.1.4	Managed Floats (Most Common Today)	17
4.2	The BOP and Interest Rates	17
4.2.1	Interest Rates → BOP	17
4.2.2	BOP → Interest Rates	18
4.3	The BOP and Inflation Rates	18
4.3.1	Imports as Anti-Inflation Tool	18
4.3.2	The Policy Dilemma	18
5	Trade Balances and Exchange Rates	18
5.1	The Theory: Devaluation Improves Trade Balance	18
5.2	Need to fix coding or just add image from text	19
5.3	The J-Curve Adjustment Path	19
5.3.1	Stage 1: Currency Contract Period (Months 0-3)	19
5.3.2	Stage 2: Pass-Through Period (Months 3-9)	20
5.3.3	Stage 3: Quantity Adjustment Period (Months 9+)	20
5.3.4	The Marshall-Lerner Condition	20
5.4	Trade Balance Adjustment: The Equation	21
5.5	Real-World Complications	21
5.5.1	“Expenditure Switching” May Not Occur	21
5.5.2	Global Supply Chains	21
5.5.3	Commodity Dependence	22
6	Capital Mobility	22
6.1	Current Account versus Financial Account Capital Flows	22
6.1.1	Current Account Capital Flows	22
6.1.2	Financial Account Capital Flows	22
6.2	Historical Patterns of Capital Mobility	22
6.2.1	Classical Gold Standard (1870-1914)	23
6.2.2	Interwar Years (1923-1938)	23
6.2.3	Fixed Exchange Rates - Bretton Woods (1944-1973)	23
6.2.4	Floating Exchange Rates (1973-1997)	23
6.2.5	The Emerging Era (1997-Present)	24
6.3	Capital Controls	24
6.3.1	Types of Capital Controls	24
6.3.2	Purposes of Capital Controls	24
6.3.3	Effectiveness of Capital Controls	25
6.4	Capital Flight	25
6.4.1	Mechanisms of Capital Flight	25
6.4.2	Famous Capital Flight Episodes	26
6.4.3	Policy Responses	26
6.5	Globalization of Capital Flows - Benefits and Risks	26
6.5.1	Benefits of Capital Mobility	26
6.5.2	Risks of Capital Mobility	26

6.5.3	The Current Debate	27
7	Conclusion and Synthesis	27
7.1	Major Themes Recap	27
7.1.1	The Balance of Payments as an Analytical Tool	27
7.1.2	The Structure of the BOP	27
7.1.3	Exchange Rates and the BOP	28
7.1.4	The J-Curve	28
7.1.5	Capital Mobility - Evolution and Implications	28
7.2	Connecting to the Real World	28
7.2.1	Why This Matters for Business	28
7.2.2	Why This Matters for Policy	29
7.2.3	Current Events Through BOP Lens	29
7.3	Key Takeaways	29
8	Questions to test understanding	30
8.1	Practical Application	30
9	Final Thoughts	30
9.1	The Big Picture	30
9.2	Developing Your BOP Intuition	31
9.3	Success Strategies	31
9.4	Office Hours and Help	31
10	Appendix: Additional Resources	32
10.1	Recommended Readings	32
10.2	Data Sources	32
10.3	Practice Problems Guidance	32

1 Introduction

Random Quote: **A student said to his master: “You teach me fighting, but you talk about peace. How do you reconcile the two?” The master replied: “It is better to be a warrior in a garden than to be a gardener in a war.”**

Context Setting

In this chapter we will explore one of the most fundamental analytical tools in international finance: the Balance of Payments.

Imagine this: You’re the CEO of a major U.S. company. You just read this morning’s *Wall Street Journal* headline: “U.S. Trade Deficit Hits Record High.” Should you be concerned about what this means for your business? Should you be worried about the future? Should you hedge your foreign exchange exposure? Should you accelerate foreign investment?

To answer these questions intelligently, you need to understand the balance of payments.

Key Quote from Chapter:

“The sort of dependence that results from exchange, i.e., from commercial transactions, is a reciprocal dependence. We cannot be dependent upon a foreigner without his being dependent on us. Now, this is what constitutes the very essence of society. To sever natural interrelations is not to make oneself independent, but to isolate oneself completely.” - Frédéric Bastiat

Why the Balance of Payments Matters

The BOP matters to **three key groups**:

- 1. Business Managers and Investors** - Signals potential exchange rate changes - Indicates possible capital controls - Forecasts market potential for exports/imports - Reveals investment restrictions or opportunities
- 2. Government Policymakers** - Guides monetary and fiscal policy decisions - Influences trade policy - Affects currency intervention decisions
- 3. Financial Market Participants** - Currency traders - Fixed income investors - Equity analysts covering multinationals

Chapter Roadmap

Today we'll cover **five major sections**:

- 1. Fundamentals of BOP Accounting** - What it is, what it measures, how it works
 - 2. The Accounts of the Balance of Payments** - Current account, capital account, financial account
 - 3. BOP Impacts on Key Macroeconomic Rates** - Exchange rates, interest rates, inflation
 - 4. Trade Balances and Exchange Rates** - The J-curve and devaluation effects
 - 5. Capital Mobility** - Historical patterns, controls, and the globalization of capital
-

2 Fundamentals of BOP Accounting

2.1 What IS the Balance of Payments?

CRITICAL DEFINITION: A Balance of Payments (BOP) statement is a statement of cash flows over an interval of time more in accord with a corporate income statement, but on a cash basis, that measures **ALL** international economic transactions between the residents of a country and foreign residents over a period of time, typically a year.

2.1.1 Key Clarifications

“Residents” doesn’t just mean individuals: - Individuals - Businesses - Government entities - Any entity domiciled in the country

“International economic transactions” include: - Trade in goods (merchandise) - Trade in services (banking, consulting, tourism) - Investment flows (buying stocks, bonds, real estate abroad) - Currency transactions - Transfer payments

2.2 Common Misconceptions About the BOP

2.2.1 Misconception #1: “The BOP is a Balance Sheet”

The Balance of Payments is not a Balance Sheet

- The word “balance” suggests a balance sheet (like a company’s assets and liabilities at a point in time)
- But the BOP is actually a **FLOW STATEMENT**
- More like an income statement or cash flow statement
- Records flows over a period of time (usually a year)

Important Note: There IS a related measure called the Net International Investment Position (NIIP) that functions like a balance sheet - we’ll discuss this later.

2.2.2 Misconception #2: “The BOP Can Be in Disequilibrium”

The phrase “the balance of payments is in disequilibrium” is **technically incorrect**. The BOP **MUST** balance. Always. By definition. Why else would it be called “**Balance** of Payments”?

What people really mean by “Disequilibrium”: - A sub-account (like the current account) is in deficit or surplus - “The balance of trade is in deficit” - “The current account is in surplus” - “The BOP is in disequilibrium”

2.2.3 Misconception #3: “Debits and Credits Work Like Corporate Accounting”

Sort of, but not exactly The BOP uses double-entry bookkeeping but with **unique definitions**:

BOP Credit (+): - An event that earns foreign exchange - An **inflow** of foreign exchange - Examples: Exports of goods, foreign investment into your country

BOP Debit (—): - An event that spends foreign exchange - An **outflow** of foreign exchange - Examples: Imports of goods, domestic investment abroad

2.3 The BOP as a Flow Statement

Think of the BOP as tracking **TWO fundamental types of transactions**:

2.3.1 1: Exchange of Real Assets

- Goods (automobiles, computers, food, clothing)
- Services (banking, consulting, tourism, education)
- Can be exchanged for:
 - Other goods and services (barter - rare today)
 - Money (the usual case)

2.3.2 2: Exchange of Financial Assets

- Stocks
- Bonds
- Direct investment in companies
- Real estate
- Bank deposits
- Can be exchanged for:
 - Other financial assets
 - Money

2.3.3 The “Follow the Cash Flow” Rule

When in doubt about BOP accounting, ask yourself:

1. Which direction did the money flow?
2. Did money flow INTO the country? → Credit (+)
3. Did money flow OUT OF the country? → Debit (–)

Example Walk-Through:

Let's classify some transactions:

Transaction: A U.S. company exports tractors to Brazil and receives payment. - **Classification:** Current account (goods), CREDIT (+) - **Why?** Money flows INTO the U.S.

Transaction: A Japanese company buys U.S. Treasury bonds. - **Classification:** Financial account, CREDIT (+)
- **Why?** Money (capital) flows INTO the U.S.

Transaction: An American tourist pays for a hotel in Paris. - **Classification:** Current account (services - travel), DEBIT (–) - **Why?** Money flows OUT OF the U.S.

Transaction: A U.S. pension fund buys shares in a German company. - **Classification:** Financial account, DEBIT (–) - **Why?** Money (capital) flows OUT OF the U.S.

2.4 The BOP Identity

At the highest level, the BOP can be summarized with this equation:

$$\text{Current Account} + \text{Capital Account} + \text{Financial Account} + \text{Net Errors and Omissions} + \text{Reserves} = 0$$

This MUST equal zero. If it doesn't, there's been an accounting error.

2.4.1 Why the Need for “Net Errors and Omissions”?

The Problem: Double-entry bookkeeping in theory vs. practice

- **In theory:** Every transaction creates TWO entries that offset
 - Export of goods (+) matched by import of currency (–)
- **In practice:** Transactions recorded independently
 - Current account transactions recorded when goods cross borders
 - Financial transactions recorded when payments clear
 - Different agencies, different timing, different methods

Result: Statistical discrepancies arise. The “net errors and omissions” account makes the BOP balance.

3 The Accounts of the Balance of Payments

Overview of the Main Accounts

The BOP has **three major sub-accounts**:

1. **Current Account** - Tracks trade and income flows occurring within the year
2. **Capital Account** - Tracks capital transfers and non-produced assets (usually small)
3. **Financial Account** - Tracks investment flows (direct investment, portfolio, other)

Plus two “balancing” accounts: - **Net Errors and Omissions** - Statistical adjustment - **Official Reserves** - Government foreign exchange holdings

Let's examine each in detail.

3.1 The Current Account

Definition: Includes all international economic transactions with income or payment flows occurring **within the year** (the current period).

3.1.1 The Four Components of the Current Account

3.1.1.1 Goods Trade (Merchandise Trade)

What it includes: - Physical products: cars, electronics, food, clothing, machinery - The oldest form of international economic activity - Most visible component

The Balance of Trade (BOT): - Goods exports MINUS goods imports - Most quoted BOP statistic in business press - “Trade surplus” or “trade deficit”

U.S. Example: Let me show you the U.S. pattern:

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.



What drives goods trade? - **Imports:** Domestic income levels, economic growth, relative prices
- **Exports:** Foreign income levels, foreign economic growth, relative prices - **Exchange rates:**
Affects relative prices

3.1.1.2 Services Trade

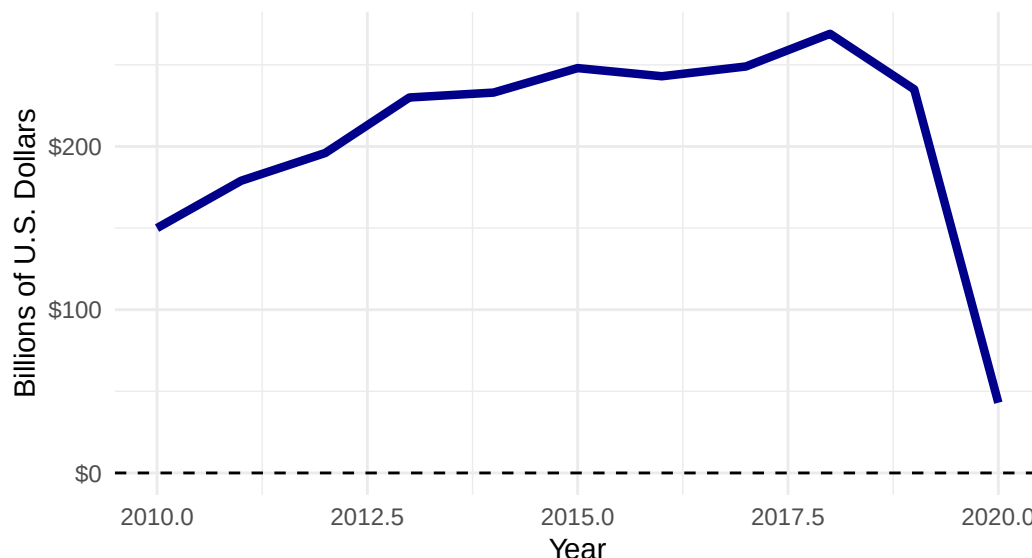
What it includes: - Financial services (banking fees, insurance) - Transportation (shipping, airline tickets) - Travel and tourism - Business services (consulting, legal, accounting) - Telecommunications
- Royalties and license fees - Education (international students)

Key Point: This is the **fastest-growing component** for developed economies.

Why? Modern economies increasingly service-based, services increasingly tradeable (digitization).

U.S. Example:

U.S. Balance on Services Trade, 2010–2020
Consistent Surplus (except 2020 pandemic effect)



Notable: U.S. consistently runs services surplus. This partially offsets goods deficit.

3.1.1.3 Primary Income

What it includes: - **Investment income:** Dividends and interest from previous investments - A U.S. company earned profits from its UK subsidiary last year → pays dividend to U.S. parent this year - **Compensation of employees:** Wages paid to non-resident workers - A Mexican citizen works in the U.S., sends wages home

Critical Distinction: - The **investment** happened in a previous period (recorded in financial account then) - The **income** from that investment happens now (recorded in current account)

Example: In 2015, Apple invested \$500 million building a manufacturing plant in China (financial account outflow in 2015). In 2023, that plant generates \$50 million in profits that Apple repatriates to the U.S. (current account inflow in 2023).

3.1.1.4 Secondary Income (Current Transfers)

What it includes: - Government grants and aid - Personal remittances (migrant workers sending money home) - Gifts - Pensions - Humanitarian aid

Key Characteristic: These are **one-way transfers**. No quid pro quo. Not payment for goods, services, or assets.

Important Topic - Global Remittances:

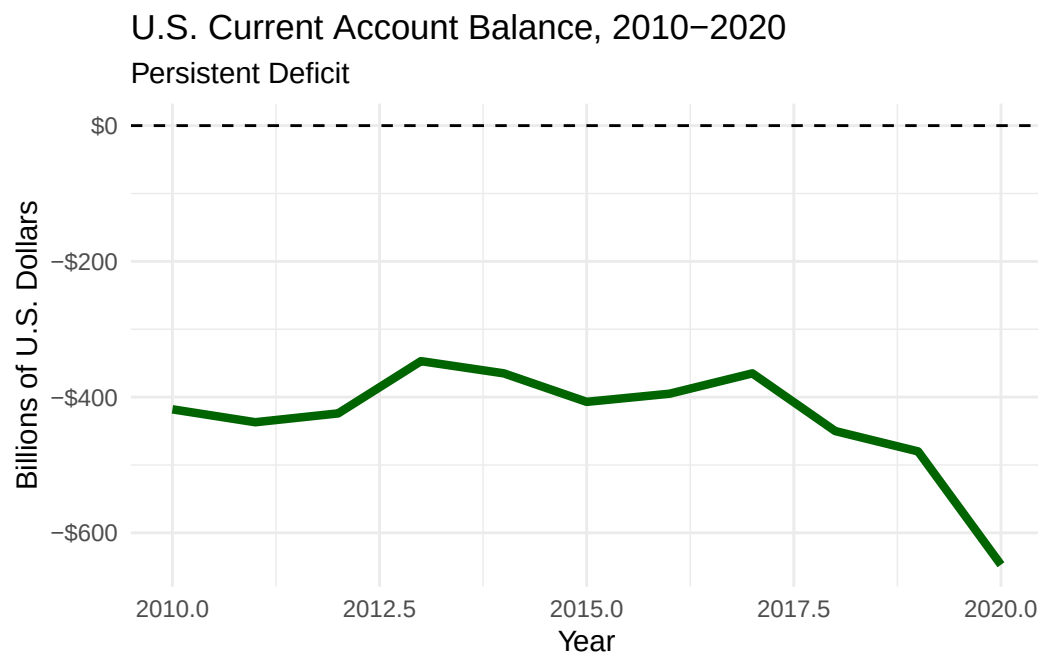
The mini-case in your textbook discusses this. Remittances have become **HUGE**: - \$700+ billion globally in 2020 - For many countries, exceeds foreign direct investment - Critical for economic development - For some countries, 20-30% of GDP!

Example: Philippines received \$35 billion in remittances in 2020 - money sent home by Filipino workers abroad. This is their **second-largest source of foreign exchange** after exports.

3.1.2 Putting It Together: The Current Account Balance

$$\text{Current Account} = (\text{Goods}) + (\text{Services}) + (\text{Primary Income}) + (\text{Secondary Income})$$

U.S. Current Account - The Full Picture:



What does this tell us? - U.S. consistently spends more abroad than it earns from abroad - This represents **either**: - Borrowing from abroad (increasing liabilities) - Selling assets to foreigners - Drawing down reserves

We'll see how this connects to the financial account shortly.

3.2 The Capital Account

Quick Note: This is relatively small and often gets lumped with the financial account in discussions.

What it includes: - Transfers of non-produced, non-financial assets - Debt forgiveness - Migrants' transfers of assets when moving

Example: A family immigrates from India to the U.S., bringing \$50,000 in savings. This is a capital account transaction.

Magnitude: Usually billions (small) compared to trillions in current and financial accounts.

3.3 The Financial Account

This is where it gets interesting. The financial account measures investment flows.

3.3.1 The Three Main Components

3.3.1.1 Direct Investment (FDI)

Definition: Investment that involves a **long-term commitment** and **significant control** over assets.

Key Threshold: Minimum 10% ownership stake.

Examples of FDI Inflow to U.S. (Credit): - Toyota builds a manufacturing plant in Kentucky - Shell Oil (Dutch company) buys a refinery in Texas - A Chinese company purchases 15% stake in a U.S. tech startup

Examples of FDI Outflow from U.S. (Debit): - Ford builds an assembly plant in Mexico - Google opens offices and data centers in Singapore - A U.S. private equity firm acquires a controlling stake in a French company

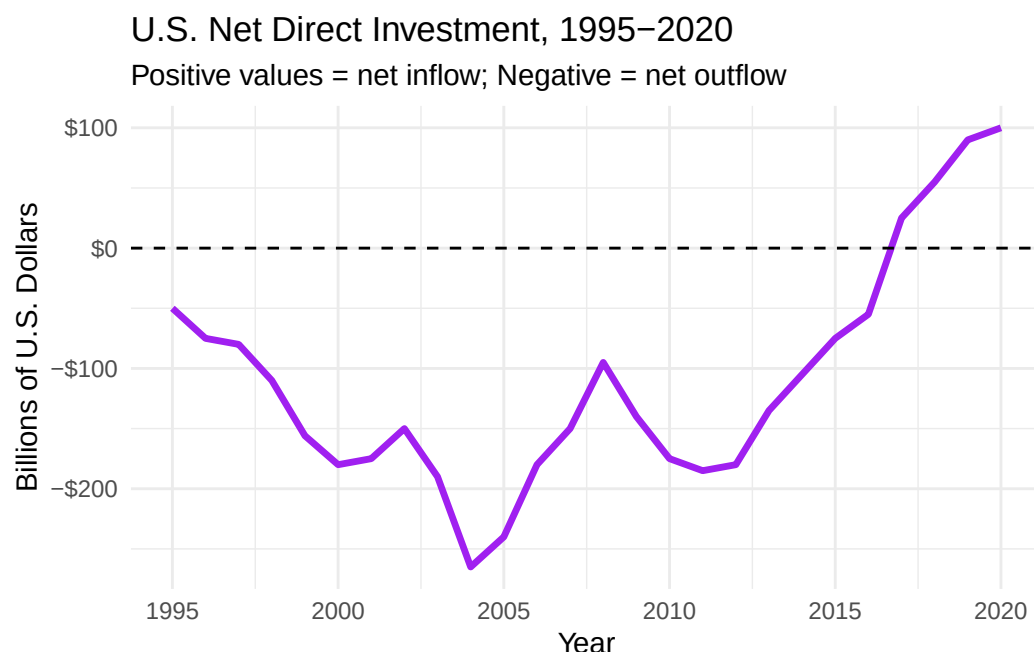
Why FDI Matters: - Creates jobs (usually) - Transfers technology and expertise - Longer-term commitment (harder to reverse quickly) - Generally viewed positively by host countries

Controversial Aspects: - Foreign control of domestic assets - Who receives the profits? - National security concerns

Historical Example: In the 1980s, Japanese companies (Sony, Honda, Toyota) made large FDI in the U.S. This created political controversy. “Are we selling America?”

Modern Parallel: Chinese FDI in the U.S. faces even greater scrutiny today due to national security concerns.

Net Direct Investment - U.S. Example:



Pattern: U.S. companies investing abroad (outflows) generally exceeded foreign companies investing in U.S. (inflows) for most years, but this has reversed recently.

3.3.1.2 Portfolio Investment

Definition: Investment in financial assets (stocks, bonds) **without significant control**.

Key Distinction from FDI: - Typically less than 10% ownership - Motivated by return, not control - Short-term oriented - Easily reversible (“hot money”)

Examples of Portfolio Inflow to U.S. (Credit): - A German pension fund buys U.S. Treasury bonds - A Japanese investor purchases 100 shares of Apple stock - A Chinese insurance company invests in U.S. corporate bonds

Examples of Portfolio Outflow from U.S. (Debit): - A U.S. mutual fund buys Mexican government bonds - An American individual invests in European stocks - A U.S. pension fund purchases bonds issued by an Australian company

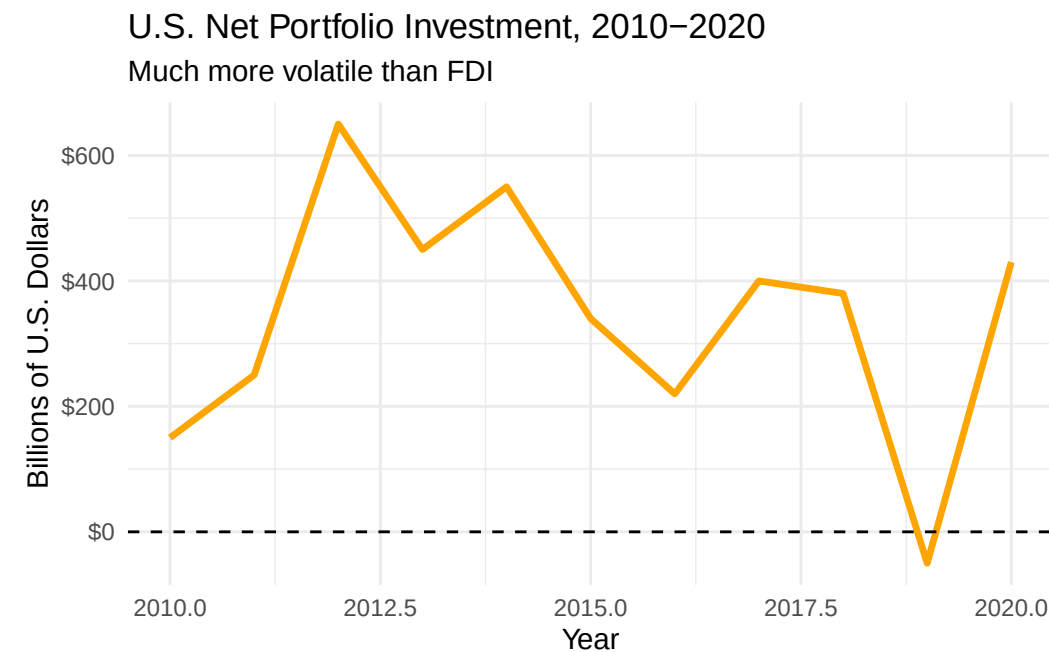
Why Portfolio Investment Matters: - MUCH larger in volume than FDI - Can move VERY quickly (contributing to financial crises) - Drives exchange rates in the short run - Connects national financial markets

The “Hot Money” Problem:

Portfolio capital can flee a country **rapidly** during a crisis: - 1997-98 Asian Financial Crisis - 2001 Argentine Crisis
- 2008-09 Global Financial Crisis - 2013 “Taper Tantrum”

Example - Asian Crisis: In early 1997, billions of dollars of portfolio capital flowed INTO Thailand, Malaysia, Indonesia. By late 1997, tens of billions flowed OUT in a matter of weeks. Currencies collapsed.

U.S. Net Portfolio Investment:



Key Observation: Much more volatile year-to-year than FDI. This is “hot money.”

3.3.1.3 Other Asset Investment

What it includes: - Trade credits - Bank loans and deposits - Currency holdings - Other receivables and payables

Example: A U.S. exporter extends 90-day credit to a Brazilian importer. This is recorded here.

3.3.2 The Fundamental Relationship: Current Account Financial Account

This is CRITICAL to understand:

In a well-functioning economy with flexible exchange rates:

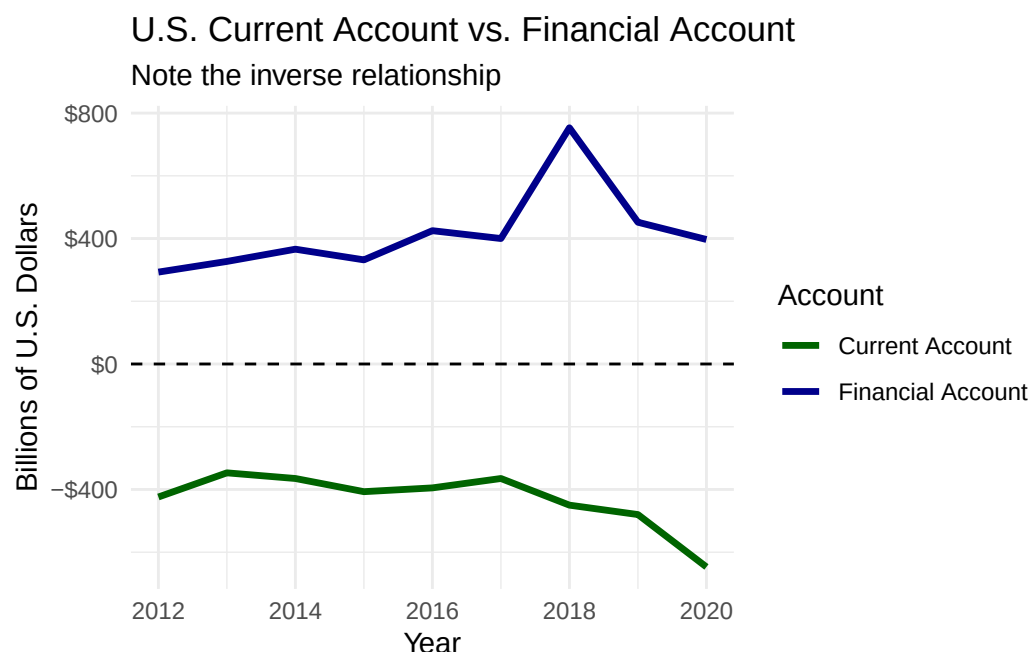
$$\text{Current Account Balance} \approx -(\text{Financial Account Balance})$$

In words: - A current account **deficit** is typically matched by a financial account **surplus** - A current account **surplus** is typically matched by a financial account **deficit**

Why? The BOP must balance! If you're spending more abroad than you're earning (current account deficit), you must be either: 1. Borrowing from abroad, or 2. Selling assets to foreigners

Both of these are financial account inflows (surplus).

U.S. Example - The Inverse Relationship:



See the pattern? When current account is negative (below zero), financial account is positive (above zero).

What does this mean? The U.S. has been running large current account deficits. How is this financed? By foreigners investing in U.S. assets (buying Treasury bonds, stocks, real estate, companies).

3.3.3 Breaking the Rules: China's Twin Surpluses

Normally impossible, but China managed it for years.

The Typical Pattern: - Current account surplus → financial account deficit (capital outflow) -
Current account deficit → financial account surplus (capital inflow)

China (2006-2013): - Current account: SURPLUS (earning more abroad than spending) -
Financial account: ALSO SURPLUS (capital flowing in)

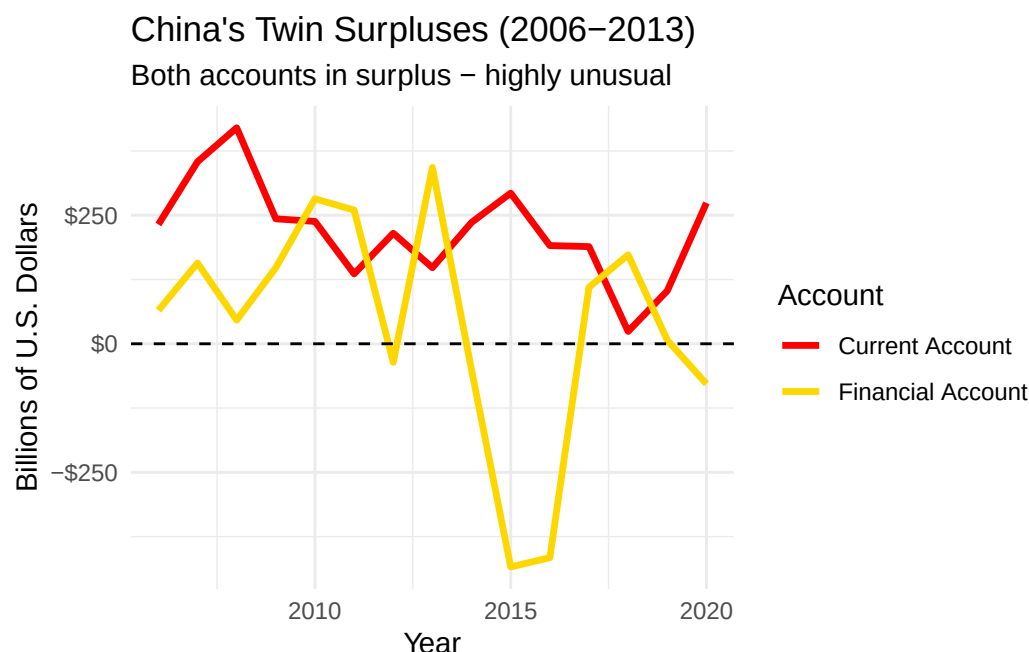
How was this possible?

1. **Massive export success** → current account surplus
2. **Attractive investment destination** → FDI inflows
3. **Strict capital controls** → prevented money from leaving
4. **Currency intervention** → government soaked up excess dollars

Result: China accumulated **\$4 trillion** in foreign exchange reserves.

The Impossible Trinity in Action: China demonstrated that with: - Fixed exchange rate (to the dollar) - Capital controls - Independent monetary policy

You CAN temporarily have surpluses in both accounts. But this is **not sustainable** long-term without controls.



Notice: Starting around 2014, pattern shifts toward “normal” - current surplus with financial deficit.

3.4 Net Errors and Omissions

Purpose: Make the BOP actually balance.

Why needed? - Data collected by different agencies - Different timing - Different methodologies
- Some transactions unreported (illegal trade, money laundering) - Honest mistakes

Typical magnitude: Tens of billions for large economies like the U.S.

Red flag: If net errors and omissions becomes **very large**, suggests: - Major capital flight - Significant illegal activity - Serious data collection problems

3.5 Official Reserves

Definition: Foreign currency assets held by a country’s central bank.

What’s included: - Major foreign currencies (dollars, euros, yen) - Gold - Special Drawing Rights (SDRs) from IMF

Role depends on exchange rate regime:

Fixed Exchange Rate: - Reserves are **critical** - Used to defend the peg - Can run out! (forces devaluation)

Floating Exchange Rate: - Reserves less critical - Used for occasional intervention - “War chest” for emergencies

Example - China’s Reserves:

At peak (2014): Nearly **\$4 trillion**

This allowed China to: - Defend the yuan if needed - Weather financial crises - Project economic power

4 BOP Impacts on Key Macroeconomic Rates

The BOP both **influences** and **is influenced by** three key macroeconomic variables: 1. Exchange rates 2. Interest rates 3. Inflation rates

4.1 The BOP and Exchange Rates

4.1.1 The Simplified BOP Equation

$$(X - M) + (CI - CO) + (FI - FO) + FXB = BOP = 0$$

Where: - X = Exports, M = Imports (Current Account) - CI = Capital inflows, CO = Capital outflows (Capital Account) - FI = Financial inflows, FO = Financial outflows (Financial Account) - FXB = Change in reserves - BOP = Balance of Payments (must equal zero)

How this affects exchange rates depends on the exchange rate regime:

4.1.2 Fixed Exchange Rate Countries

Government's Responsibility: Keep $BOP \approx 0$ by intervening in FX markets.

Scenario 1: Current + Capital + Financial Accounts > 0 (Surplus) - Excess DEMAND for domestic currency - Domestic currency wants to APPRECIATE - To prevent appreciation and maintain fixed rate: - Government SELLS domestic currency - BUYS foreign currency (increases reserves)

Scenario 2: Current + Capital + Financial Accounts < 0 (Deficit) - Excess SUPPLY of domestic currency - Domestic currency wants to DEPRECIATE - To prevent depreciation and maintain fixed rate: - Government BUYS domestic currency - SELLS foreign currency (decreases reserves)

Critical Problem: Reserves can run out! Then forced to devalue.

Historical Example - British Pound (1992): - UK trying to maintain pound's value in European Exchange Rate Mechanism - Current + financial accounts in deficit - Bank of England spent billions buying pounds - Reserves depleted - Forced to abandon fixed rate - Pound fell 15% in a day - Cost UK taxpayers billions

The "Speculator" Story: George Soros famously bet against the pound, made \$1 billion in profit. Was he evil? Or just recognizing the inevitable?

4.1.3 Floating Exchange Rate Countries

Government's Responsibility: None! Let the market determine the rate.

Scenario: Current + Capital + Financial Accounts < 0 (Deficit) - Excess SUPPLY of domestic currency on world markets - Like any good in excess supply, price falls - Currency DEPRECIATES automatically - This (eventually) corrects the imbalance: - Depreciation makes exports cheaper → exports ↑ - Depreciation makes imports more expensive → imports ↓ - Current account improves

Key Point: In theory, floating rates are **self-correcting**. No reserves needed.

Reality: Doesn't always work smoothly or quickly (we'll discuss J-curve shortly).

4.1.4 Managed Floats (Most Common Today)

What most countries actually do: - Claim to have "floating" rate - But intervene occasionally to: - Smooth volatility - Prevent "overshooting" - Maintain competitiveness

Tool: Change interest rates to influence capital flows.

Example: Country's currency depreciating too fast: - Raise domestic interest rates - This attracts foreign capital (portfolio investment) - Increased capital inflow supports currency - Currency stabilizes

Trade-off: Higher interest rates slow domestic economy. Political cost.

4.2 The BOP and Interest Rates

Two-way relationship:

4.2.1 Interest Rates → BOP

Relatively HIGH domestic interest rates: - Attract foreign portfolio investment - Financial account SURPLUS (capital inflows) - Supports currency value - **But** raises domestic borrowing costs

Relatively LOW domestic interest rates: - Encourage capital to seek higher returns abroad - Financial account DEFICIT (capital outflows) - Weakens currency value - **But** stimulates domestic economy

4.2.2 BOP → Interest Rates

Large current account deficit financed by foreign investment: - Foreigners buying domestic bonds keeps interest rates LOW - U.S. example: Despite huge deficits and government borrowing, U.S. rates stayed low (until recently) - Why? Foreign demand for U.S. Treasuries (China, Japan, oil exporters)

Current account surplus: - Country exporting capital - Can lead to LOW domestic interest rates (excess savings) - Japan example: Decades of current account surplus, near-zero interest rates

4.3 The BOP and Inflation Rates

4.3.1 Imports as Anti-Inflation Tool

Mechanism: - Import of low-priced goods puts ceiling on domestic prices - Foreign competition limits what domestic firms can charge - Greater imports → lower inflation (all else equal)

Example - Walmart Effect: - Walmart sources globally, offers low prices - Forces competitors to lower prices - Benefits consumers (lower prices) - But costs jobs in domestic manufacturing

Trade-off: - Lower inflation - **But** lower GDP and employment (if imports replace domestic production)

4.3.2 The Policy Dilemma

Countries must balance: - Benefits of open trade (lower inflation, more consumer choice) - Costs of open trade (job losses, industry decline)

Political Reality: The benefits are diffuse (all consumers benefit slightly), costs are concentrated (specific workers/industries hurt badly). This creates political pressure for protectionism.

5 Trade Balances and Exchange Rates

5.1 The Theory: Devaluation Improves Trade Balance

Simple Logic: 1. Country devalues its currency (or currency depreciates in floating regime) 2. Exports become cheaper for foreign buyers → export quantity ↑ 3. Imports become more expensive for domestic buyers → import quantity ↓ 4. Trade balance improves

Sounds straightforward, right? Not so fast!

5.2 Need to fix coding or just add image from text

5.3 The J-Curve Adjustment Path

Reality is more complex. The trade balance adjustment follows three stages:

5.3.1 Stage 1: Currency Contract Period (Months 0-3)

What happens: - Currency suddenly devalues/depreciates - BUT existing contracts already in place - Exporters and importers must fulfill obligations

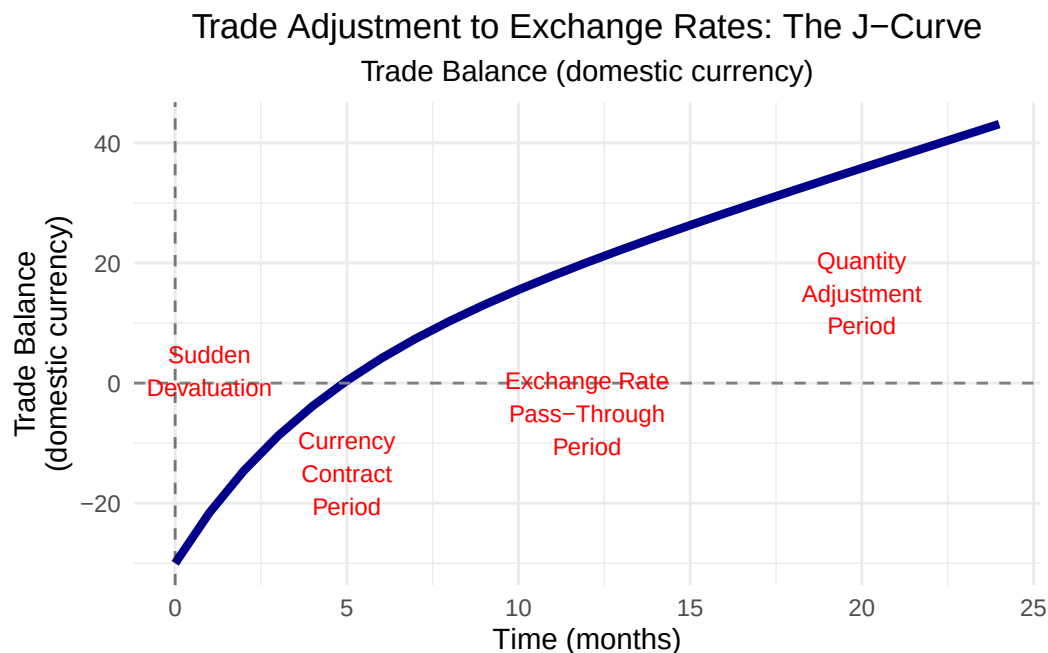
Assume: U.S. dollar suddenly depreciates 20%

Impact on U.S. trade balance: - **Exports:** Most contracts denominated in dollars → revenue unchanged

- **Imports:** Many contracts denominated in foreign currency → dollar cost INCREASES - **Net effect:** Trade balance gets **WORSE!**

Example: - Before: U.S. imports €100 million of German goods, €1 = \$1.20, cost = \$120 million - After 20% dollar depreciation: €1 = \$1.44, same €100 million now costs \$144 million - U.S. import bill increased by \$24 million - Exports unchanged (contracts already signed in dollars) - Trade deficit larger!

Graphical Representation:



See the “J” shape? Gets worse before it gets better.

5.3.2 Stage 2: Pass-Through Period (Months 3-9)

What happens: - Old contracts expire, new ones negotiated - Exchange rate changes “pass through” to prices - Foreign exporters to U.S. raise dollar prices (to cover costs in their currency) - U.S. exporters can now price more competitively in foreign markets

U.S. import prices: ↑ (foreign goods more expensive in dollars) **U.S. export prices:** ↓ (U.S. goods cheaper in foreign currency)

Critical Question: Do foreign exporters pass through the FULL exchange rate change?

Often NO!

Example - Japanese Auto Exports to U.S.: - Yen appreciates 20% vs. dollar - In theory, Japanese cars should be 20% more expensive in U.S. - In practice, Japanese manufacturers often raise prices only 5-10% - They absorb some of the exchange rate change (lower profit margins) - Why? To maintain market share

This is called “pricing to market” - very common with manufactured goods.

5.3.3 Stage 3: Quantity Adjustment Period (Months 9+)

What happens: - Prices have adjusted - Consumers respond to new prices - Import demand decreases (more expensive) - Export demand increases (cheaper) - Trade balance IMPROVES

Critical Factors:

Price Elasticity of Demand: - How responsive is quantity demanded to price changes? - High elasticity → big quantity response → trade balance improves significantly - Low elasticity → small quantity response → trade balance barely improves

Example:

High elasticity: Consumer electronics - Dollar depreciates 20% - Korean TVs now 20% more expensive in U.S. - U.S. consumers buy fewer Korean TVs, more U.S.-made alternatives - Import quantity drops significantly

Low elasticity: Oil - Dollar depreciates 20% - Oil 20% more expensive in dollars - U.S. still needs oil (few alternatives in short run) - Import quantity barely changes - Import **bill** actually increases (higher price × similar quantity)

5.3.4 The Marshall-Lerner Condition

For devaluation to improve the trade balance:

$$|\varepsilon_X| + |\varepsilon_M| > 1$$

Where: - ε_X = price elasticity of demand for exports - ε_M = price elasticity of demand for imports

In words: The sum of elasticities must exceed 1.

If this condition is NOT met: Devaluation can worsen the trade balance even in the long run!

Empirical Evidence: For most industrialized countries, condition is met, but adjustment takes 12-24 months.

5.4 Trade Balance Adjustment: The Equation

Let's formalize this mathematically.

Trade Balance in domestic currency:

$$\text{Trade Balance} = (P_X^{\$} \times Q_X) - (S^{\$/fc} \times P_M^{fc} \times Q_M)$$

Where: - $P_X^{\$}$ = price of exports in dollars - Q_X = quantity of exports - $S^{\$/fc}$ = spot exchange rate (dollars per foreign currency) - P_M^{fc} = price of imports in foreign currency - Q_M = quantity of imports

Immediate impact of devaluation (Stage 1): - $S^{\$/fc}$ increases (takes more dollars to buy foreign currency) - Q_X and Q_M unchanged (contracts already signed) - Import bill increases - Trade balance worsens

Later impact (Stage 3): - Q_X increases (exports cheaper to foreigners) - Q_M decreases (imports more expensive to domestic buyers) - If elasticities are high enough, quantity effects dominate price effects - Trade balance improves

5.5 Real-World Complications

5.5.1 "Expenditure Switching" May Not Occur

Problem: Even if imports become expensive, domestic alternatives may not exist.

Example: U.S. manufacturing of certain goods has disappeared: - Consumer electronics - Textiles and apparel - Some machinery

Even if Chinese imports become 30% more expensive, Americans may have to keep buying them (no U.S.-made alternative).

5.5.2 Global Supply Chains

Modern reality: Most products are made with inputs from multiple countries.

Example: iPhone - Designed in California - Assembled in China - Components from Japan, Korea, Taiwan, Germany - Software from U.S.

When the dollar depreciates: - Yes, iPhones assembled in China become cheaper for Europeans - **But** many components are bought in dollars (U.S.-designed chips) - Net effect on trade balance is ambiguous

5.5.3 Commodity Dependence

If country heavily depends on imported commodities (oil, food): - Demand is price inelastic
- Devaluation increases import bill - May not improve trade balance much (or at all)

Example - India: - Imports 80% of oil - Rupee depreciates - Oil bill increases dramatically - Trade balance may worsen despite cheaper exports

6 Capital Mobility

6.1 Current Account versus Financial Account Capital Flows

We need to distinguish between **two types** of international capital flows:

6.1.1 Current Account Capital Flows

Associated with trade in goods and services: - Payment for exports/imports - Trade credits - Related financial transactions

Characteristics: - Directly linked to “real” economic activity - Relatively stable - Somewhat predictable

6.1.2 Financial Account Capital Flows

Associated with investment: - Portfolio investment (stocks, bonds) - Direct investment (FDI) - Bank loans - Currency speculation

Characteristics: - Can be massive in magnitude - Can reverse very quickly (“hot money”) - Driven by expectations, sentiment, risk appetite - Can cause financial crises

Key Point: Financial account flows now **dwarf** current account flows. The financial tail wags the economic dog!

Magnitude Example: - Daily global trade in goods: ~\$50 billion - Daily global foreign exchange trading: ~\$6.6 **trillion** - Financial flows are 100+ times larger!

6.2 Historical Patterns of Capital Mobility

Capital has **not** always moved freely. The degree of capital mobility has varied dramatically over time.

6.2.1 Classical Gold Standard (1870-1914)

Characteristics: - **Fixed exchange rates** (currencies tied to gold) - **Increasing capital mobility** (for major powers) - **Trade dominated capital flows**

Environment: - British Empire at peak - Pax Britannica - relatively stable political environment
- Capital flowed to finance trade, infrastructure (railroads) - Colonies and developing countries received investment

Limitations: - Primarily flows among industrialized nations - Many developing countries had limited access - System dependent on gold convertibility

6.2.2 Interwar Years (1923-1938)

Characteristics: - **Protectionism and nationalism** - **Severely restricted capital mobility** - **Trade barriers erected**

Environment: - Great Depression - Competitive devaluations (“beggar thy neighbor”) - Capital controls widespread - International trade collapsed

Result: Economic disaster. Demonstrated the dangers of closing economies.

6.2.3 Fixed Exchange Rates - Bretton Woods (1944-1973)

Characteristics: - **Fixed exchange rates** (currencies pegged to dollar, dollar to gold) - **Free trade encouraged** (GATT) - **Capital controls permitted** (even encouraged)

Key Distinction: - Current account transactions: Free movement of capital - Capital account transactions: Subject to controls

Philosophy: - International trade benefits everyone - International capital flows can be destabilizing
- Countries should be able to control capital flows

Why it worked: - Economies recovering from WWII - U.S. economically dominant - Capital controls effective (limited financial innovation)

Why it ended: - Eurocurrency markets grew - Capital controls became porous - Speculation against fixed rates intensified - System collapsed 1971-1973

6.2.4 Floating Exchange Rates (1973-1997)

Characteristics: - **Major currencies floating** (dollar, pound, yen, mark) - **Emerging markets:** Varied regimes - **Increasing capital mobility** for developed countries - **Emerging markets:** More cautious

Environment: - Financial innovation (derivatives, securitization) - Deregulation and liberalization
- Eurocurrency markets expand - Emerging markets start opening up

Divergence: - **Developed countries:** Embrace capital mobility - **Emerging markets:** More gradual, often maintaining controls

Result: Growing integration of developed country financial markets.

Ended with: Asian Financial Crisis (1997-1998)

6.2.5 The Emerging Era (1997-Present)

Characteristics: - **Emerging markets** increasingly important - **Massive capital flows** to emerging markets - **Periodic crises** and capital flow reversals - **Renewed debate** about capital controls

Major Events: - **1997-98:** Asian Financial Crisis - **2001:** Argentine Crisis - **2008-09:** Global Financial Crisis - **2010-12:** European Sovereign Debt Crisis - **2013:** “Taper Tantrum” - **2015-16:** China capital outflows - **2020:** COVID-19 pandemic

Current State: - Capital can move faster and in greater volume than ever - But many countries maintain controls - “Hot money” flows create volatility - Debate about optimal degree of capital openness

6.3 Capital Controls

Definition: Any restriction that limits or alters the rate or direction of capital movement into or out of a country.

6.3.1 Types of Capital Controls

Outflow Controls: Restrictions on residents investing abroad or moving money out - Require approval for foreign investment - Taxes on capital outflows - Limits on amount that can be transferred - Prohibition on certain transactions

Inflow Controls: Restrictions on foreigners investing in the country or types of investment allowed - Taxes on capital inflows (Chile’s encaje) - Required minimum holding periods - Restrictions on sectors (foreigners cannot own media, real estate, etc.) - Limits on foreign ownership percentages

6.3.2 Purposes of Capital Controls

See Exhibit 3.8 in your textbook for comprehensive list. **Key purposes:**

1. Prevent Currency Appreciation (Chile 1990s) - Problem: Capital flooding in - Currency appreciating - Hurting export competitiveness - Solution: Tax on short-term capital inflows (encaje) - Required percentage of inflow to be deposited interest-free - Effectively a tax, higher for shorter-term investments

2. Prevent Capital Flight (Malaysia 1998) - Problem: Asian Financial Crisis, capital fleeing - Currency collapsing - Solution: Temporarily freeze capital outflows - Allowed trade-related flows - Blocked financial flows - Stabilized currency

3. Maintain Monetary Policy Independence - Problem: Impossible Trinity - Can’t have: fixed exchange rate + capital mobility + independent monetary policy - Solution: Control capital to keep other two

4. Protect Domestic Industries - Restrict foreign ownership - National security concerns - Strategic industries

6.3.3 Effectiveness of Capital Controls

Debate continues:

Arguments FOR controls: - Can provide breathing room during crisis - Allow countries to maintain preferred policies - Prevent speculative attacks - Protect sovereignty

Arguments AGAINST controls: - Reduce efficiency of capital allocation - Create opportunities for corruption - Can be evaded (particularly in modern financial system) - May signal weakness, trigger capital flight - Reduce foreign investment

Empirical Evidence: Mixed

- Can be effective short-term
- Less effective long-term
- Controls on inflows more acceptable than controls on outflows
- Comprehensive controls more effective than partial

IMF Position (evolved): - Previously: Capital controls generally bad - Now: May be useful tool in some circumstances - Should be temporary - Should be part of broader policy package

6.4 Capital Flight

Definition: Rapid outflow of capital in opposition to or in fear of domestic political and economic conditions and policies.

Characteristics: - Sudden and large - Driven by fear/panic, not return optimization - Often illegal or semi-legal

6.4.1 Mechanisms of Capital Flight

1. Legal Transfers - Normal banking system - Wire transfers - Convert to foreign currency, move abroad

2. Physical Currency Transfer - Literally smuggling cash - Hidden in luggage, containers - Risky, costly, but happens

3. False Invoicing - Over-invoice imports: Pay \$150 for \$100 of imports, \$50 goes to foreign account - **Under-invoice exports:** Sell \$100 of exports for \$50, keep \$50 abroad

Example: Company in Argentina (fearing peso devaluation): - Imports \$1 million of machinery - Bills show \$1.5 million - Pays \$1.5 million (sends money abroad) - \$1 million goes to real supplier - \$0.5 million deposited in company's Swiss account

4. Cryptocurrency (New) - Convert local currency to Bitcoin/crypto - Transfer crypto across borders - Convert to foreign currency abroad - Increasingly used, increasingly tracked by authorities

6.4.2 Famous Capital Flight Episodes

Argentina (2001) - Banks froze accounts - People desperate to get money out - Massive capital flight beforehand contributed to crisis

Venezuela (2000s-present) - Political instability, economic collapse - Those who could move money out - Billions fled - Worsened economic crisis

Russia (2014-2015) - Ukraine crisis, sanctions - Ruble under pressure - Capital flight - Contributed to recession

6.4.3 Policy Responses

Preventative: - Maintain sound economic policies - Build credibility - Avoid policies that encourage flight

Reactive: - Capital controls (Malaysia 1998) - Bank freezes (Argentina 2001) - Currency restrictions

Effectiveness: Once capital flight begins, very hard to stop. Prevention much better than cure.

6.5 Globalization of Capital Flows - Benefits and Risks

6.5.1 Benefits of Capital Mobility

1. Efficient Capital Allocation - Capital flows to highest return projects globally - Not constrained by domestic savings

2. Risk Diversification - Investors can diversify globally - Countries can share risks

3. Access to Capital - Developing countries can access foreign savings - Finance development faster

4. Discipline on Governments - Bad policies punished by capital outflows - Provides market discipline

6.5.2 Risks of Capital Mobility

1. Volatility and “Hot Money” - Capital can reverse quickly - Creates boom-bust cycles

2. Contagion - Problems in one country spread to others - “When the U.S. sneezes, emerging markets catch pneumonia”

3. Loss of Policy Autonomy - Impossible Trinity constraints - Market forces may override domestic preferences

4. Inequality - Benefits of globalization unequally distributed - Some win big, some lose

6.5.3 The Current Debate

After 2008 Financial Crisis, even the IMF reconsidered:

“Notwithstanding these benefits [of capital mobility], many EMEs [emerging market economies] are concerned that the recent surge in capital inflows could cause problems... capital controls are again in the news.”

The New Consensus: - Capital mobility has benefits - But unfettered capital flows can be destabilizing - Temporary, targeted capital controls may be appropriate - As part of broader policy package - Not a substitute for good policies

Especially relevant for: - Large, sudden inflows (appreciation pressure) - Highly volatile flows (portfolio “hot money”) - When other policy tools exhausted

7 Conclusion and Synthesis

7.1 Major Themes Recap

7.1.1 The Balance of Payments as an Analytical Tool

What we learned: - The BOP is a comprehensive record of ALL international transactions - It's a **flow statement**, not a balance sheet - It **MUST** balance (by definition) - Sub-accounts can be in surplus or deficit - Provides critical signals about: - Currency pressure - Policy effectiveness - Economic health - International position

7.1.2 The Structure of the BOP

Three major accounts:

- 1. Current Account:** - Goods, services, income, transfers - Most visible (trade balance) - Surplus/deficit has implications for: - Currency value - Domestic production - Employment
- 2. Capital Account:** - Small, often overlooked - Capital transfers, non-produced assets
- 3. Financial Account:** - Direct investment (FDI) - Portfolio investment
- Other investments - Increasingly dominant - Can dwarf current account in magnitude

Fundamental Relationship: - Current account deficit → Financial account surplus (borrowing/selling assets) - Current account surplus → Financial account deficit (lending/buying assets) - Except when governments intervene heavily (China example)

7.1.3 Exchange Rates and the BOP

The interaction depends on exchange rate regime:

Fixed Rates: - Government must intervene with reserves - Deficit draws down reserves (can run out!) - Surplus builds reserves

Floating Rates: - Market determines rate - Deficits → depreciation → eventually corrects - Surpluses → appreciation → eventually corrects

But adjustment is NOT instantaneous!

7.1.4 The J-Curve

Critical insight: - Currency depreciation/devaluation doesn't immediately improve trade balance - Initially makes it **worse** - Eventually improves (if elasticities are adequate) - Timing: 12-24 months for full adjustment

Three stages: 1. Currency contract period: Worse 2. Pass-through period: Still worse, but stabilizing 3. Quantity adjustment period: Improvement

Policy implication: Devaluation is not a quick fix. Patience required.

7.1.5 Capital Mobility - Evolution and Implications

Historical perspective: - Capital mobility has varied dramatically over time - Not a one-way trend toward openness - Depends on political-economic philosophy of the era

Current era: - Unprecedented capital mobility (developed countries) - Emerging markets more varied - Financial flows »> trade flows - Can be destabilizing

The Debate: - Benefits: Efficiency, growth, diversification - Costs: Volatility, crisis, inequality - No consensus on optimal degree of openness - Context-dependent

7.2 Connecting to the Real World

7.2.1 Why This Matters for Business

1. Currency Exposure Management - BOP trends signal currency movements - Current account deficit + capital outflows = depreciation pressure - Time to hedge?

2. Market Assessment - Is a country's growth sustainable? - Current account deficit financed by FDI (stable) or portfolio flows (volatile)?

3. Policy Anticipation - BOP problems often lead to: - Capital controls - Currency restrictions - Import barriers - Can affect your operations

Example: You're a U.S. company exporting to Brazil. Brazil's current account deficit is widening, reserves are falling, portfolio capital is fleeing. **What do you do?** - Tighten credit terms - Demand payment in dollars (not reais) - Hedge receivables - Consider reducing exposure

7.2.2 Why This Matters for Policy

National policy dilemmas:

1. **The Impossible Trinity** - Can't have all three: fixed rate, capital mobility, monetary independence - Must choose two - BOP data reveals which countries are struggling with this
2. **Trade-offs** - Open trade → lower inflation, more choice - But also → job displacement, industry decline - BOP shows these trade-offs in action
3. **Financial Stability vs. Growth** - Open capital markets → access to funds, growth - But also → vulnerability to crises - BOP data can signal building imbalances

7.2.3 Current Events Through BOP Lens

Let's apply what we learned to current situations:

U.S. Current Account Deficit: - Persistent and large (\$647 billion in 2020) - Financed by foreign purchases of U.S. assets - Is this sustainable? What could go wrong? - What if foreigners lose appetite for U.S. assets?

China's Current Account Surplus: - Still large (\$274 billion in 2020) - But financial account now often in deficit (capital leaving) - Indicates less confidence in Chinese economy? - Government using reserves to stabilize

Emerging Markets' Challenge: - Many running current account deficits - Dependent on portfolio inflows - Vulnerable to Fed policy changes - "Taper tantrum" risk

7.3 Key Takeaways

5 Critical Points:

1. **The BOP Always Balances** - It's an identity - If someone says "BOP disequilibrium," they mean a sub-account
2. **Current and Financial Accounts Are (Usually) Inverse** - Deficit in one, surplus in the other - This is normal and necessary - Exceptions require government intervention
3. **Adjustment to Exchange Rate Changes Takes Time** - J-curve is real - 12-24 months typical - Don't expect quick fixes
4. **Capital Flows Can Dwarf Trade Flows** - Financial account often larger than current account - Can cause currency volatility - Can trigger crises
5. **Capital Mobility Is a Double-Edged Sword** - Benefits: efficiency, growth, diversification - Costs: volatility, crisis, loss of autonomy - Optimal degree is debated

8 Questions to test understanding

Question 1: “If Germany is running a large current account surplus, what must be true about its financial account (assuming no major change in reserves)?”

Expected Answer: Financial account must be in deficit. Germans are investing/lending abroad more than foreigners are investing in Germany.

Question 2: “A country’s currency suddenly depreciates 25%. Will its trade balance immediately improve?”

Expected Answer: No! J-curve effect. Trade balance will likely **worsen** initially, improve later.

Question 3: “China accumulated \$4 trillion in reserves. How did this happen given standard BOP relationships?”

Expected Answer: Twin surpluses (current + financial). Only possible with capital controls and currency intervention.

Question 4: “Why might capital controls be more acceptable on inflows than outflows?”

Expected Answer: - Inflow controls don’t trap people’s money - Can slow appreciation, “hot money” - Outflow controls signal desperation - Trap people’s savings - Often evaded

8.1 Practical Application

Scenario: You’re analyzing whether to invest in a Brazilian mining company. You notice: - Brazil’s current account deficit is widening - Portfolio capital is flowing out - Reserves are declining - Real (currency) is under pressure

Questions: - What does this tell you about Brazil’s economy? - What are the risks to your investment? - What might the Brazilian government do?

9 Final Thoughts

9.1 The Big Picture

The balance of payments is more than just accounting. It’s a window into: - How countries interact economically - How capital flows globally - How exchange rates adjust
- How crises develop

Every business decision with an international dimension involves BOP concepts: - Should we hedge? - Should we invest abroad? - Where should we raise capital? - Which markets should we enter?

Every policy debate about globalization involves BOP issues: - Trade deficits - Capital flows - Exchange rates - Financial crises

9.2 Developing Your BOP Intuition

To really master this material:

1. **Follow the Data** - Check BOP statistics for countries you're interested in - IMF publishes comprehensive data - BEA (for U.S.) - Look for patterns and changes
2. **Read the News Through a BOP Lens** - When you read about currency movements, ask: "What BOP factors are driving this?" - Trade disputes: Current account - Capital flows: Financial account - Currency intervention: Reserves
3. **Think About Mechanisms** - Don't just memorize relationships - Understand **why** current account and financial account are inverse - Understand **why** depreciation has a J-curve effect - The "why" sticks better than the "what"
4. **Connect to Earlier Material** - Impossible Trinity (Chapter 2) → Explains BOP dynamics under different regimes - Comparative advantage (Chapter 1) → Explains current account patterns - Everything connects

9.3 Success Strategies

For the exam: - Practice the calculations (percentage changes, trade balances) - Understand the accounts structure

- Know the J-curve stages - Be able to explain relationships (not just state them)

For your career: - This is **fundamental** knowledge for international finance/business - BOP analysis is used daily by: - Currency traders - International portfolio managers - Corporate treasurers - Economic analysts - Consultants - Government policymakers

For life: - Understanding BOP helps you make sense of international economic news - Affects your investments - Affects policy debates you'll vote on

9.4 Office Hours and Help

I'm here to help you master this material: - Office hours: [specify times] - Email: [specify] - Happy to discuss concepts - Happy to review problems - Happy to discuss current events

Don't hesitate to email.

Thank you for your attention today. See you next class!

10 Appendix: Additional Resources

10.1 Recommended Readings

For deeper understanding:

1. **IMF Balance of Payments Manual** (technical but authoritative)
2. **The Economist** - Weekly coverage of BOP issues globally
3. **Financial Times** - Excellent international economics coverage
4. **Council on Foreign Relations** - BOP explainers and policy debates

10.2 Data Sources

Where to find BOP data:

1. **IMF International Financial Statistics**
 - <https://data.imf.org>
 - Comprehensive, standardized across countries
2. **U.S. Bureau of Economic Analysis**
 - <https://www.bea.gov>
 - Detailed U.S. data, historical series
3. **Trading Economics**
 - <https://tradingeconomics.com>
 - User-friendly interface, charts
4. **OECD Data**
 - <https://data.oecd.org>
 - Focus on developed countries

10.3 Practice Problems Guidance

For Problems 3.1-3.4 (Australia): - Calculate each component balance - Note patterns over time - What was happening in Australian economy?

For Problems 3.5-3.9 (India): - Compare to Australia
- Different pattern - why? - What explains India's deficits?

For Problems 3.10-3.14 (China): - Look for twin surpluses - When did pattern change? - What explains the change?

Work through these carefully. They reinforce concepts.